

Software Engineering Lab 3 – Architectural Justification

Faculty Research Grant & Publication Tracker

Name: AaroHi Jaiswal SRN:PES1UG24CS009

Section: A

Architecture Selection

We chose Layered Architecture for the Faculty Research Grant & Publication Tracker System.

Architectural Choice

The system is organized into three layers: Presentation, Business Logic, and Data. The Presentation Layer contains the Web Application (User Interface); the Business Logic Layer contains Authentication & Authorization, Grant Management, Publication Management, Publication Approval, and Research Monitoring & Analytics services; and the Data Layer contains the Research Database.

Reason 1 – Separation of Concerns

The system has distinct responsibilities such as grant management, publication management, co-author approval, and research analytics. Separating these responsibilities into layers makes the system easier to understand, maintain, test, and modify without unnecessarily affecting other layers.

Reason 2 – Centralized Data Management

The system manages related grant, expense, publication, approval, user, and analytics information. A dedicated Data Layer provides a centralized mechanism for storing and retrieving this information while keeping data management separate from application services.

Security Advantage

Authentication and Authorization is separated as a business service so access can be controlled according to user roles. Faculty Researchers, Co-authors, and Research Deans can be given appropriate permissions, helping protect grant, publication, approval, and analytics information from unauthorized access.

Performance Benefit

The layered structure allows database access and business operations to be optimized independently. Frequently used operations such as grant balance calculation, publication retrieval, approval processing, and analytics data access can therefore be handled efficiently under normal system usage.